ETP story formalization · majority voting at temperature 0.7

Stories generated from Equational Theories Project implications (`storyform.py`), formalized back by each model over OpenRouter, and graded syntactically (`checkform.py`). Correct = exact, or an accepted symmetry (sides swapped, or both equations uniformly dualized).

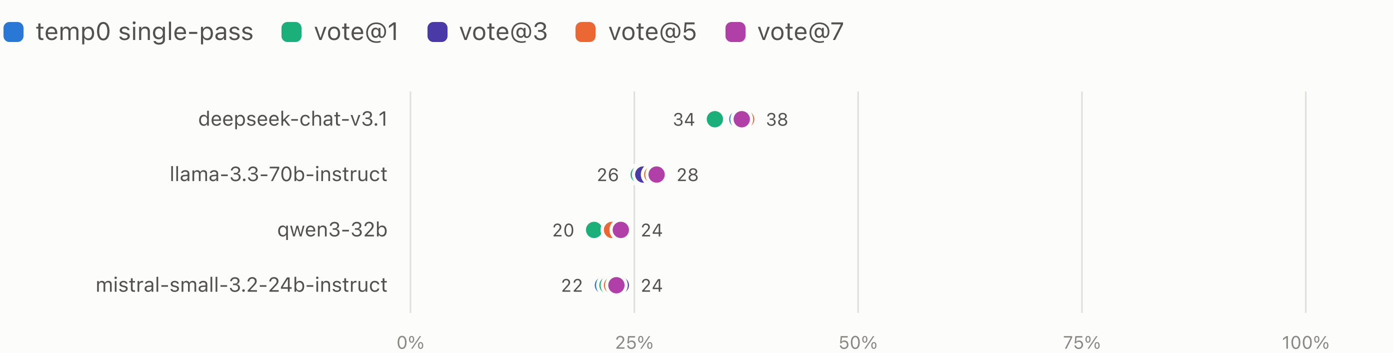
experiments/11-majority-voting/runs/run-eqstrat20-s0-story-t0.7-t0-baseline · experiments/11-majority-voting/runs/run-eqstrat20-s0-story-t0.7-vote1 · experiments/11-majority-voting/runs/run-eqstrat20-s0-story-t0.7-vote3 · experiments/11-majority-voting/runs/run-eqstrat20-s0-story-t0.7-vote5 · experiments/11-majority-voting/runs/run-eqstrat20-s0-story-t0.7-vote7

Runs	Models	Stories	Graded calls	Overall correct
5	4	200	4,000	27%

200 pairs, 20 per per-equation operation bin 1–10, both laws from the same bin · seed 0 · temp0 single-pass vs vote@1 vs vote@3 vs vote@5 vs vote@7

FIGURE 1 **Correct rate by model and condition — balanced 200**

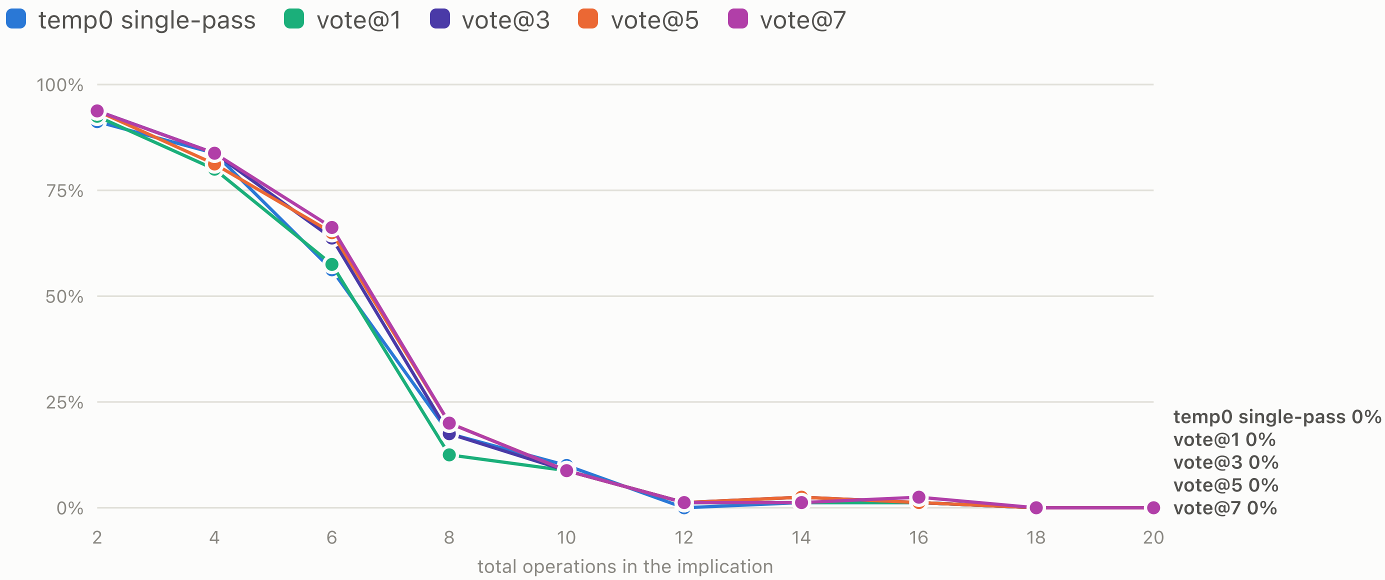
The same 200 stories under each condition; each row compares one model across all conditions. Sample: 200 pairs, 20 per per-equation operation bin 1–10 — both laws of a pair carry the bin's exact operation count.



▶ Data: correct rate per model and condition

FIGURE 2 **Accuracy vs operation count — balanced 200**

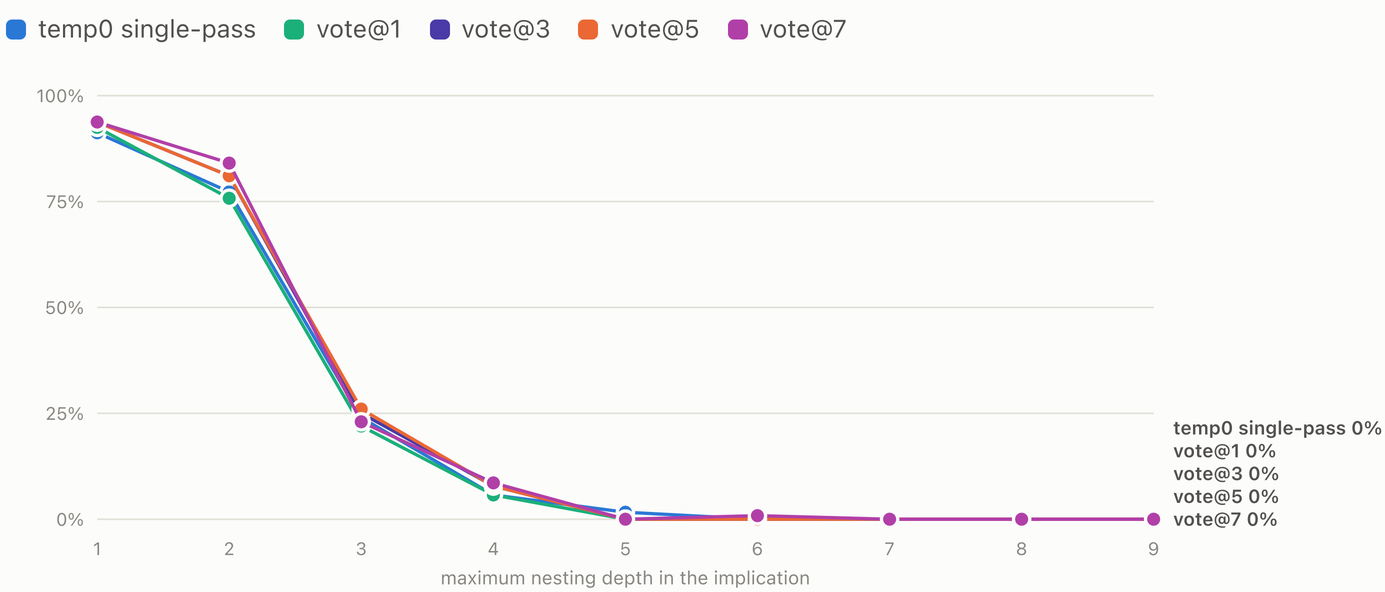
Correct rate pooled over all models, by the implication's total operation count (both equations combined). Sample: 200 pairs, 20 per per-equation operation bin 1–10 — both laws of a pair carry the bin's exact operation count.



▶ Data: pooled correct rate per operation-count bin

FIGURE 3 **Accuracy vs nesting depth — balanced 200**

Correct rate pooled over all models, by the deepest operation nesting across the four equation sides. Sample: 200 pairs, 20 per per-equation operation bin 1–10 — both laws of a pair carry the bin's exact operation count.

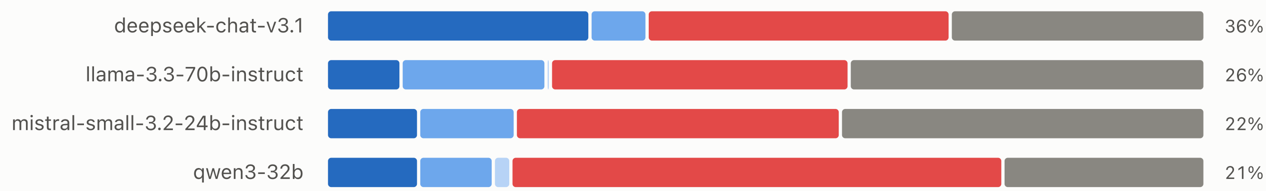


▶ Data: pooled correct rate per depth bin

FIGURE 4 **Verdict composition — temp0 single-pass · balanced 200**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 200 pairs, 20 per per-equation operation bin 1–10 — both laws of a pair carry the bin's exact operation count.

exact · correct-swapped · correct-dualized · wrong · unparseable

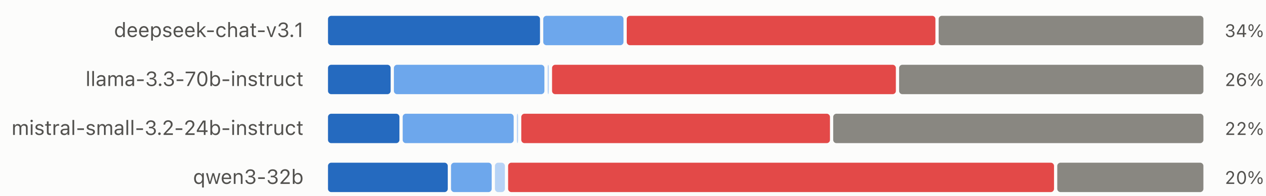


▶ Data: verdict counts per model

FIGURE 5 **Verdict composition — vote@1 · balanced 200**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 200 pairs, 20 per per-equation operation bin 1–10 — both laws of a pair carry the bin's exact operation count.

exact · correct-swapped · correct-dualized · wrong · unparseable

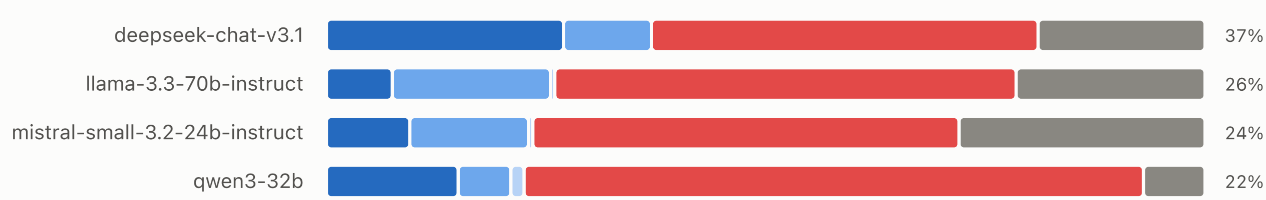


▶ Data: verdict counts per model

FIGURE 6 **Verdict composition — vote@3 · balanced 200**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 200 pairs, 20 per per-equation operation bin 1–10 — both laws of a pair carry the bin's exact operation count.

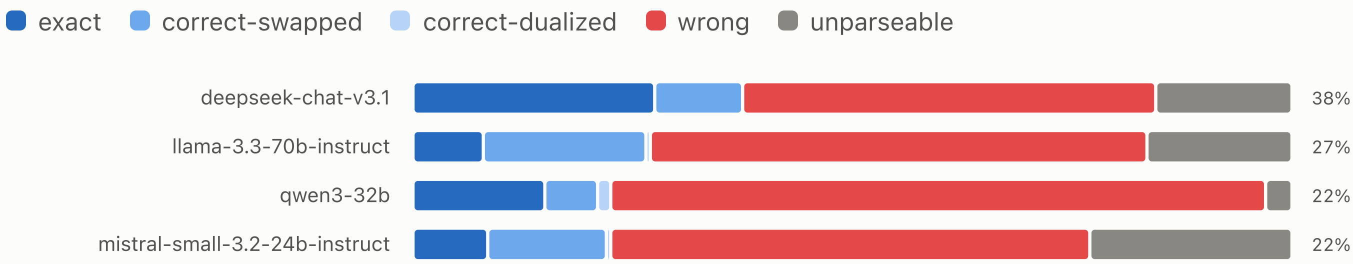
exact · correct-swapped · correct-dualized · wrong · unparseable



▶ Data: verdict counts per model

FIGURE 7 **Verdict composition — vote@5 · balanced 200**

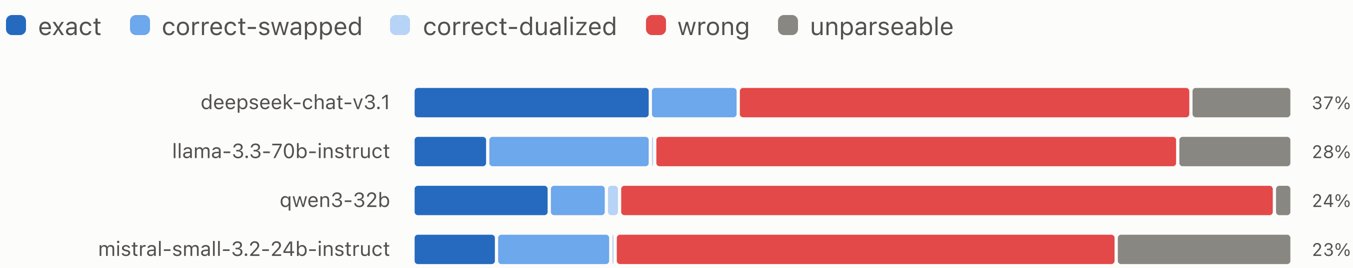
How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 200 pairs, 20 per per-equation operation bin 1–10 — both laws of a pair carry the bin's exact operation count.



▶ Data: verdict counts per model

FIGURE 8 **Verdict composition — vote@7 · balanced 200**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 200 pairs, 20 per per-equation operation bin 1–10 — both laws of a pair carry the bin's exact operation count.



▶ Data: verdict counts per model